



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 03.08.2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No : PHA G616
Course Title : Pharmaceutical Administration & Management (PAM)
Instructor-in-Charge : Ankit Jain (ankit.j@pilani.bits-pilani.ac.in, 3170-M)

- 1. Course Description:** This course introduces students to various organizational aspects, focusing on pharmaceutical management. Topics include technology innovation, drug planning, strategy, project execution, product development, production, scale-up, marketing, IPR, HR, industrial relations, documentation, R&D, and ethics.
- 2. Scope and Objective of the Course:** The objective of the course is to help students understand new drug and product considerations and planning; strategic considerations; project implementation; product development; production management & scale-up; marketing strategy; human resources development; ethical issues; and R&D management.
- 3. Course learning outcomes:**
CLO1: Explain the principles of pharmaceutical product innovation, new drug and product planning, strategic decision-making, project implementation, and product development processes in the pharmaceutical industry.
CLO2: Apply the concepts of production management, scale-up, documentation, intellectual property rights (IPR), and marketing strategies to evaluate pharmaceutical product development and commercialization.
CLO3: Analyze the roles of research and development (R&D) management, human resource development, industrial relations, and ethical practices in achieving sustainable and compliant pharmaceutical operations.
- 4. Textbooks and references:**

TB1. O.P. Khanna, Industrial Engineering and Management, Dhanpat Rai Publication, Revised Edition 2015

TB2. Sanford Bolton, Charles Bon, Pharmaceutical Statistics, Informa Healthcare, 5th Edition, 2010.

Reference Books: Students should refer to the following books for further reading.

RB1. Raja, B. Smarta: *Strategic Pharmaceutical Marketing*, PharmaMed Press, Second Edition 2017.

RB2. Mickey C. Smith, *Principles of Pharmaceutical Marketing*, Lea & Febiger, 3rd Edition.

5. Course Format & Weekly Workflow

Wk no.	Module No.	Description	Reference	Learning outcomes
I	1. Introduction	L1 Introduction to management	-	Scope & objective of the course
	2. Management of Operations	L2.1 Introduction- Definition of product, production, and its management, its objectives, and benefits	T1-Ch. 2 T1 Ch. 4 T1 Ch. 7	Understanding of product, production, and their management.
		L2.2 Production Management- its components and measurement of productivity		



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II		L2.3 Pharmaceutical Plant Layout	Class Notes	Factors dictating the selection of a particular production type require a detailed understanding of each aspect of production. How to select an appropriate location for a new plant and plan its layout.
		L2.4 Make or buy decision and EBQ determination		
		L2.5 Details about each production aspect		
III	3. Sales forecasting	L2.6 Line of balance technique	Class notes	Estimating future market demand by using one or a combination of techniques.
		L3.1-2 Different techniques for making forecasts of estimated sales		
IV	4. Management of inventory	L4.1-3 Different techniques, Models of inventory management, Economic order quantity, ERP, and MRP systems	T1-Ch. 24	Classification of items in inventory, ways of ordering new materials, and software to manage inventories
V- VI	5. Project Planning	L5.1-4 Network analysis-CPM and PERT	Class notes	Learn the technique of management of the production cycle in a time-bound manner and meet deadlines
VI	6. Pricing policy	L6.1. Pricing Policy of Pharmaceuticals	DPCO (available online)	How pricing of drugs regulated by the NPPA
VI- VII	7. Product Costing, Break-Even Analysis	L7.1-3. Elements of costing, overheads, calculations on the cost of man and materials, depreciation, and breakeven analysis	T1-Ch.27	Basic Financial Management, Significance of fixed and variable costs
VII - IX	8. Marketing Management	L8.1 Types of marketing concepts, sales vs. marketing, market research, market planning.	T1, Ch. 31 T1, Ch. 28 R1, Ch. 2 Class Notes	Significance of marketing, designing of a marketing mix, product positioning, and brand value.
		L8.2 Budget, Marketing audit. Decision on new products, Channels of distribution		
		L8.3 Stretching of product line. Advertising		
		L8.4 Branding and labeling, differences between FMCG and pharmaceuticals marketing practices		
		L8. 5-6 Innovation and creativity in pharmaceutical marketing		
		L8. 7-8 New drugs and key factors to launch them successfully in the market		
X	9. Strategic Management	L9.1-2 Options in the Indian environment, Segmentation & product positioning	T1, Ch. 31 R1, Ch. 4 Class notes	4Ps and product life cycle



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X- XI	10. R&D Management	L10.1-2 Managing research teams & new product	R1, Ch. 2	Factors affecting new product design and estimation of resources required
XI	11. HRD	L11.1. Activities of the HR department, welfare, and factories act	T1, Ch. 20 Class notes	Planning of human resources for organizational effectiveness
	12. Ethical Aspects	L12.1 Code of conduct, Insidious trading, social responsibility, PEST factors	T1, Ch. 38 Class notes	Concept of professional & business ethics
XII - XII I	13. Research Methodology	L13. 1-3 Introduction to different aspects of research methodology, literature review, hypothesis, designing of experiments, data collection and compilation, and different study designs	T2, Ch. 4 Class notes	Importance of Research Design, Effect of internal validity and external validity on research design
		L13. 4. Optimization techniques using factorial designs	Class notes	Technique to obtain the optimized product on varying process parameters with a minimum no. of experiments

6. Evaluation Scheme:

Evaluation Component ^{§#*}	Weightage (%)	Duration	Date & Time	Nature of components
Continuous assessments (Quizzes, assignments, etc.)	30% (7.5% for each quiz 15% for seminar)	TBA	Quizzes during the lecture hours For the seminar (TBA)	CB+OB
Mid-Sem Exam	30	90 min.	As announced in the Timetable	OB
Comprehensive Exam	40	180 min.	As announced in the Timetable	CB

[§]Students are strongly advised to prepare their own notes based on class lectures and relevant information from the textbook and reference material, as only these notes will be allowed for consultation during assessments of open-book evaluation components. Photocopies of any material, written or printed, will not be permitted. Stapled sheets or loose sheets of information, written or printed, will not be allowed.

^{*}Slides used during class hours provide key information for which additional supportive information is expected to be collected from sources aforementioned. Recent developments in the area/topic will be discussed in class, and such discussions will be considered a primary source of information for assessments.

[#]Quizzes may be conducted as part of the evaluation component at random during contact hours, with/without prior intimation, and sometimes outside class contact hours. Hence, it is expected that students come prepared to every class on topics covered in earlier contact hours.

Grading would be done using the bunching procedure. In borderline cases, subjective judgment will be used to award the grades. Awarding all eight grades (i.e., from A to E) is not mandatory. Subjective judgment based on attendance at lectures and tutorials, participation in quizzes, students' involvement in the course, and performance in the lab component would be used to award grades. The student shall not be considered "exposed" to the course unless he/she demonstrates appreciable skill in both the laboratory and theory components of the course and through classroom participation. Attending tutorials and appearing in surprise quizzes are very important. Students who have not participated in quizzes, submitted assignments, or completed any tests/labs/comprehensive exams will not be considered



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as having completed/exposed to the course. Students are also expected to attend classes regularly to gain proper exposure and to evaluate their knowledge.

Criterion for NC: 30% of the average of the top three performers or 40% of the median, whichever is lower, or a suitable modification at the discretion of the IC with the concurrence of the Department.

- 7. Reading Assignments:** Students are advised to read and collect additional information on the above-mentioned topics as per the given schedule.
- 8. Attendance Policy:** Each student is expected to attend all classes and participate in all evaluation components.
- 9. Chamber Consultation Hours:** To be announced in class.
- 10. Make-up policy:** Makeup will be given only for genuine cases (accompanied by proof) for the Mid-semester or comprehensive exam only.
- 11. Notices:** All notices concerning the course will be emailed to the students.

Instructor-in-charge
PHA G616